VBCUA: Work Completed and 8-Week Plan

Voice-Based Concept Understanding Analyser · Vedulapurapu Sai Pavan Kumar · 10 October 2026

Summary

VBCUA is a working, end-to-end prototype. A student records an explanation of a concept; the system transcribes it, scores conceptual understanding and speech fluency, writes AI feedback, saves the attempt and exports a PDF report. It was designed, built and documented solo in June 2026.

Next 8 weeks: turn the prototype into a short research paper. Fix the scoring method, collect a small human-rated set of spoken explanations, and measure how well the automated scores agree with human judgement and how speech-recognition errors affect them.

Work completed

Problem addressed

Students preparing for viva voce, presentations and technical interviews cannot easily tell whether their spoken explanation of a concept is correct and complete, and faculty cannot give every student detailed oral feedback.

What was built

Module	What it does	Technology
Audio input	Record in the browser or upload .wav / .mp3 / .m4a; waveform preview	Streamlit, Librosa, Matplotlib
Speech-to-text	Transcribes the explanation; cloud model with a local fallback	Google Gemini, OpenAI Whisper
Semantic analysis	Compares the transcript with a reference definition; checks required keywords	Sentence-BERT (all-MiniLM-L6-v2)
Speech analysis	Filler-word rate, pause ratio, speaking volume	Librosa
Scoring	Understanding, fluency and overall score; grade Strong / Moderate / Poor	Python rules
AI feedback	Personalised strengths, gaps and suggestions; rule-based fallback without an API key	Google Gemini
Concept library	Three seeded concepts (ML, Cloud Computing, DBMS); users can add more	FastAPI, SQLAlchemy
History	Every attempt saved with transcript and sub-scores; past attempts can be reopened	PostgreSQL / SQLite
Reports	Downloadable PDF assessment report per attempt	ReportLab
Launcher	One command checks config, prepares the database, starts both services	run.py

Architecture

A Streamlit dashboard calls a FastAPI backend with 7 REST endpoints. The backend runs a five-stage pipeline (transcribe, extract audio features, score semantics, score fluency, generate feedback), stores the result and serves a PDF on request. Each stage is a separate service module, so it can be swapped or tested on its own.

Documentation produced

Stage	Documents
Ideation	Brainstorming and prioritisation, problem statement, empathy map
Requirements	Functional and non-functional requirements, customer journey map, technology stack, data-flow diagrams
Design	Problem-solution fit, proposed solution, solution architecture
Planning	Product backlog and sprint plan
Testing	Functional, performance and user-acceptance test reports
Delivery	Project report, README with setup steps, project and code demo videos

Skills demonstrated

Full-stack Python (FastAPI, Streamlit, SQLAlchemy), speech processing (Whisper, Librosa), NLP with sentence embeddings, LLM integration with fallbacks, PDF generation, and structured project documentation.

Known gaps

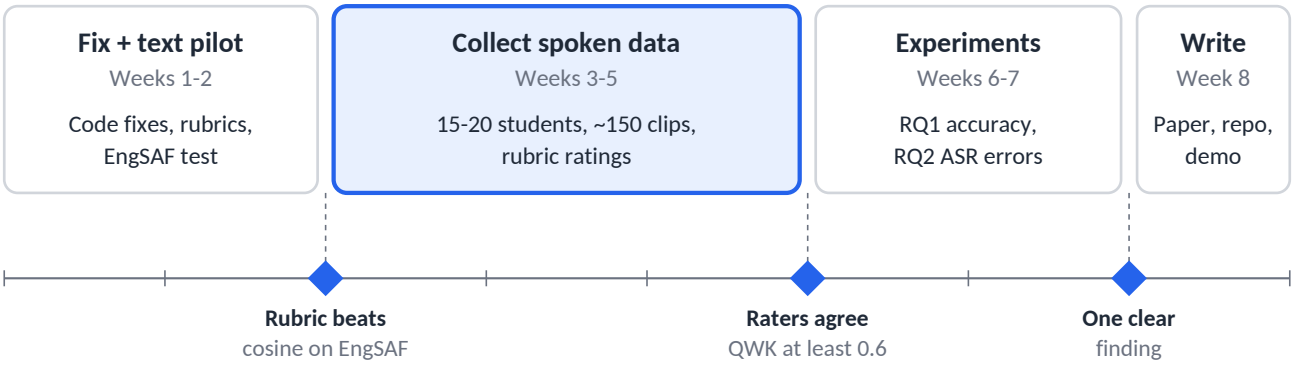
The prototype works, but five issues must be fixed before its scores can be trusted or published.

Gap	Why it matters
No human-rated data or results yet	Without comparison to human scores, accuracy cannot be claimed
Similarity score has a floor of about 50 (cosine rescaled as (cos + 1) / 2)	A fluent off-topic answer can be graded Moderate; reading the reference aloud scores Strong
Filler words counted on cleaned transcripts	Whisper and Gemini drop "um/uh", so filler counts are near zero; volume measures the microphone, not the speaker
Gemini 1.5 Flash and the google-generativeai SDK are retired	Cloud transcription and AI feedback calls fail and silently fall back
Documents describe login, extra endpoints and test results not in the code	Align them before sharing the repo with reviewers or recruiters

8-week plan

The plan answers two research questions with about 150 recorded explanations and ends with a submittable short paper in week 8. Weeks count from the day you start.

Question	What it asks
RQ1: Accuracy	How closely does rubric-based scoring of a spoken explanation agree with a human expert, compared with the current similarity score?
RQ2: Robustness	How much do speech-recognition errors on Indian-English technical speech change the scores?



Each diamond is a checkpoint: if it fails, revise that phase before moving on.

Weeks 1-2: Fix and pilot on text

- [ ] Make Whisper (small) the default transcriber; log the model name per attempt
- [ ] Move to the google-genai SDK and a current Gemini model, or an open LLM
- [ ] Remove the volume score and the (cos + 1) / 2 rescaling; show understanding and delivery as separate scores
- [ ] Store every sub-score in the database
- [ ] Correct the project documents to match the code
- [ ] Pick 6 CS concepts; write a rubric for each with a faculty member: 4-6 key points + 2-3 common misconceptions, each scored 0 / 1 / 2
- [ ] Write an evaluation script that rescores all data and reports agreement (QWK, Spearman)
- [ ] Run cosine vs rubric-LLM scoring on the public EngSAF dataset of written engineering answers

Checkpoint Rubric scoring beats cosine on EngSAF.

Weeks 3-5: Collect spoken data

- [ ] Get ethics approval and consent forms through a faculty member
- [ ] Pilot with 3 students; fix anything confusing
- [ ] Record 15-20 students × 6 concepts, 60-120 s each (about 100-150 clips) using the VBCUA app
- [ ] Manually correct 30 transcripts to measure speech-recognition error rate (WER)
- [ ] Rate all clips yourself with the rubric; a second rater scores 30% of them

Checkpoint The two raters agree (QWK ≥ 0.6); otherwise simplify the rubric to key point present / absent.

Weeks 6-7: Run experiments

- [ ] RQ1: compare the old VBCUA score, cosine to several reference answers, and rubric-LLM scoring against human scores
- [ ] RQ2: rescore with Whisper tiny / base / small and with corrected transcripts; plot score error against WER
- [ ] Make 2 figures and 2 tables

Checkpoint One clear finding to report.

Week 8: Write and submit

- [ ] Write a 4-6 page short paper: problem, dataset, method, results, limitations
- [ ] Clean the repo; release code, rubrics, transcripts and scores (audio only with consent)
- [ ] Record a 2-minute demo; update README and LinkedIn

Expected outputs and where to submit

By week 8: a validated scoring method, a small labelled dataset, a cleaned public repo and a short paper draft.

Venue	Why it fits	Typical deadline*
IEEE T4E (India)	Indian educational technology; accepts short papers	Mid-year
BEA Workshop (ACL / NAACL)	NLP for education; short and dataset papers	Feb-Mar
AIED short / late-breaking	AI in education	Late Jan - early Feb
ISCA SLaTE workshop	Speech technology in education (fits RQ2)	Varies

* Based on the 2026 cycle; confirm each 2027 call before planning around it.

After the paper (optional): test whether hesitation in speech predicts wrong answers, add adaptive follow-up viva questions, or run a feedback study with a class. Each can become a second paper.

Sources

EngSAF dataset, IIT Bombay: arxiv.org/abs/2407.12818
CrisperWhisper (Whisper drops fillers): arxiv.org/abs/2408.16589
google-generativeai deprecation: github.com/google-gemini/deprecated-generative-ai-python
AIED deadlines: mldeadlines.com/conference/aied
BEA 2026 proceedings: aclanthology.org/2026.bea-1.65